# Lecture 2

# Row Reduction and Echelon Forms

In this lecture, we will get more practice with row reduction and in the process introduce two important types of matrix forms. We will also discuss when a linear system has a unique solution, infinitely many solutions, or no solution. Lastly, we will introduce a convenient parameter called the rank of a matrix.

## 2.1 Row echelon form (REF)

Consider the linear system

$$\begin{aligned} x_1 + 5x_2 - 2x_4 - x_5 + 7x_6 &= -4 \\ 2x_2 - 2x_3 + 3x_6 &= 0 \\ -9x_4 - x_5 + x_6 &= -1 \\ 5x_5 + x_6 &= 5 \\ 0 &= 0 \end{aligned}$$

having augmented matrix

$$\begin{bmatrix} 1 & 5 & 0 & -2 & -1 & 7 & -4 \\ 0 & 2 & -2 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -9 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 5 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The above augmented matrix has the following properties:

**P1.** All nonzero rows are above any rows of all zeros.

**P2.** The leftmost nonzero entry of a row is to the right of the leftmost nonzero entry of the row above it.

Any matrix satisfying properties P1 and P2 is said to be in **row echelon form** (**REF**). In REF, the leftmost nonzero entry in a row is called a **leading entry**:

$$\begin{bmatrix} \mathbf{1} & 5 & 0 & -2 & -1 & 7 & -4 \\ 0 & \mathbf{2} & -2 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \mathbf{-9} & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & \mathbf{5} & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

A consequence of property P2 is that every entry below a leading entry is zero:

$$\begin{bmatrix} \mathbf{1} & 5 & 0 & -2 & -4 & -1 & -7 \\ \mathbf{0} & \mathbf{2} & -2 & 0 & 0 & 3 & 0 \\ \mathbf{0} & \mathbf{0} & 0 & \mathbf{-9} & -1 & 1 & -1 \\ \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & \mathbf{5} & 1 & 5 \\ \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & \mathbf{0} & 0 & 0 \end{bmatrix}$$

We can perform elementary row operations, or **row reduction**, to transform a matrix into REF.

**Example 2.1.** Explain why the following matrices are not in REF. Use elementary row operations to put them in REF.

$$\mathbf{M} = \begin{bmatrix} 3 & -1 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \end{bmatrix} \quad \mathbf{N} = \begin{bmatrix} 7 & 5 & 0 & -3 \\ 0 & 3 & -1 & 1 \\ 0 & 6 & -5 & 2 \end{bmatrix}$$

*Solution.* Matrix $\mathbf{M}$ fails property P1. To put $\mathbf{M}$ in REF we interchange $R_2$ with $R_3$:

$$\mathbf{M} = \begin{bmatrix} 3 & -1 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 3 & -1 & 0 & 3 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix $\mathbf{N}$ fails property P2. To put $\mathbf{N}$ in REF we perform the operation $-2R_2 + R_3 \to R_3$:

$$\begin{bmatrix} 7 & 5 & 0 & -3 \\ 0 & 3 & -1 & 1 \\ 0 & 6 & -5 & 2 \end{bmatrix} \xrightarrow{-2R_2+R_3} \begin{bmatrix} 7 & 5 & 0 & -3 \\ 0 & 3 & -1 & 1 \\ 0 & 0 & -3 & 0 \end{bmatrix}$$

□

Why is REF useful? Certain properties of a matrix can be easily deduced if it is in REF. For now, REF is useful to us for solving a linear system of equations. If an augmented matrix is in REF, we can use **back substitution** to solve the system, just as we did in Lecture 1. For example, consider the system

$$\begin{aligned} 8x_1 - 2x_2 + x_3 &= 4 \\ 3x_2 - x_3 &= 7 \\ 2x_3 &= 4 \end{aligned}$$

whose augmented matrix is already in REF:

$$\begin{bmatrix} 8 & -2 & 1 & 4 \\ 0 & 3 & -1 & 7 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

From the last equation we obtain that $2x_3 = 4$, and thus $x_3 = 2$. Substituting $x_3 = 2$ into the second equation we obtain that $x_2 = 3$. Substituting $x_3 = 2$ and $x_2 = 3$ into the first equation we obtain that $x_1 = 1$.

## 2.2 Reduced row echelon form (RREF)

Although REF simplifies the problem of solving a linear system, later on in the course we will need to completely row reduce matrices into what is called **reduced row echelon form (RREF)**. A matrix is in RREF if it is in REF (so it satisfies properties P1 and P2) and in addition satisfies the following properties:

**P3.** The leading entry in each nonzero row is a 1.

**P4.** All the entries above (and below) a leading 1 are all zero.

A leading 1 in the RREF of a matrix is called a **pivot**. For example, the following matrix in RREF:

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

has three pivots:

$$\begin{bmatrix} \mathbf{1} & 6 & \mathbf{0} & 3 & \mathbf{0} & 0 \\ \mathbf{0} & 0 & \mathbf{1} & -4 & \mathbf{0} & 5 \\ \mathbf{0} & 0 & \mathbf{0} & 0 & \mathbf{1} & 7 \end{bmatrix}$$

**Example 2.2.** Use row reduction to transform the matrix into RREF.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

*Solution.* The first step is to make the top leftmost entry nonzero:

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_1} \begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Now create a leading 1 in the first row:

$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \xrightarrow{\frac{1}{3}R_1} \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Create zeros under the newly created leading 1:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \xrightarrow{-3R_1+R_2} \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Create a leading 1 in the second row:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

Create zeros under the newly created leading 1:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix} \xrightarrow{-3R_2+R_3} \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

We have now completed the top-to-bottom phase of the row reduction algorithm. In the next phase, we work bottom-to-top and create zeros **above** the leading 1's. Create zeros above the leading 1 in the third row:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 1 & -3 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{-R_3+R_2} \begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 2 & 5 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{-2R_3+R_1} \begin{bmatrix} 1 & -3 & 4 & -3 & 0 & -3 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

Create zeros above the leading 1 in the second row:

$$\begin{bmatrix} 1 & -3 & 4 & -3 & 0 & -3 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \xrightarrow{3R_2+R_1} \begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

This completes the row reduction algorithm and the matrix is in RREF. □

**Example 2.3.** Use row reduction to solve the linear system.

$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 &= 8 \\ x_1 + 2x_2 + 4x_3 &= 8 \\ 3x_1 + 6x_2 + 9x_3 &= 12 \end{aligned}$$

*Solution.* The augmented matrix is

$$\begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 2 & 4 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

Create a leading 1 in the first row:

$$\begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 2 & 4 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

Create zeros under the first leading 1:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{-R_1+R_2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 4 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 4 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{-3R_1+R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The system is consistent, however, there are only 2 nonzero rows but 3 unknown variables. This means that the solution set will contain $3 - 2 = 1$ **free parameter**. The second row in the augmented matrix is equivalent to the equation:

$$x_3 = 4.$$

The first row is equivalent to the equation:

$$x_1 + 2x_2 + 3x_3 = 4$$

and after substituting $x_3 = 4$ we obtain

$$x_1 + 2x_2 = -8.$$

We now must choose one of the variables $x_1$ or $x_2$ to be a parameter, say $t$, and solve for the remaining variable. If we set $x_2 = t$ then from $x_1 + 2x_2 = -8$ we obtain that

$$x_1 = -8 - 2t.$$

We can therefore write the solution set for the linear system as

$$\begin{aligned} x_1 &= -8 - 2t \\ x_2 &= t \\ x_3 &= 4 \end{aligned} \tag{2.1}$$

**where $t$ can be any real number**. If we had chosen $x_1$ to be the parameter, say $x_1 = t$, then the solution set can be written as

$$\begin{aligned} x_1 &= t \\ x_2 &= -4 - \tfrac{1}{2}t \\ x_3 &= 4 \end{aligned} \tag{2.2}$$

Although (2.1) and (2.2) are two different parameterizations, they both give the same solution set. □

In general, if a linear system has $n$ unknown variables and the row reduced augmented matrix has $r$ leading entries, then the number of free parameters $d$ in the solution set is

$$d = n - r.$$

Thus, when performing back substitution, we will have to set $d$ of the unknown variables to arbitrary parameters. In the previous example, there are $n = 3$ unknown variables and the row reduced augmented matrix contained $r = 2$ leading entries. The number of free parameters was therefore

$$d = n - r = 3 - 2 = 1.$$

Because the number of leading entries $r$ in the row reduced coefficient matrix determine the number of free parameters, we will refer to $r$ as the **rank** of the coefficient matrix:

$$r = \text{rank}(\mathbf{A}).$$

Later in the course, we will give a more geometric interpretation to rank($\mathbf{A}$).

**Example 2.4.** Solve the linear system represented by the augmented matrix

$$\begin{bmatrix} 1 & -7 & 2 & -5 & 8 & 10 \\ 0 & 1 & -3 & 3 & 1 & -5 \\ 0 & 0 & 0 & 1 & -1 & 4 \end{bmatrix}$$

*Solution.* The number of unknowns is $n = 5$ and the augmented matrix has rank $r = 3$ (leading entries). Thus, the solution set is parameterized by $d = 5 - 3 = 2$ free variables, call them $t$ and $s$. The last equation of the augmented matrix is $x_4 - x_5 = 4$. We choose $x_5$ to be the first parameter so we set $x_5 = t$. Therefore, $x_4 = 4 + t$. The second equation of the augmented matrix is

$$x_2 - 3x_3 + 3x_4 + x_5 = -5$$

and the unassigned variables are $x_2$ and $x_3$. We choose $x_3$ to be the second parameter, say $x_3 = s$. Then

$$\begin{aligned} x_2 &= -5 + 3x_3 - 3x_4 - x_5 \\ &= -5 + 3s - 3(4 + t) - t \\ &= -17 - 4t + 3s. \end{aligned}$$

We now use the first equation of the augmented matrix to write $x_1$ in terms of the other variables:

$$\begin{aligned} x_1 &= 10 + 7x_2 - 2x_3 + 5x_4 - 8x_5 \\ &= 10 + 7(-17 - 4t + 3s) - 2s + 5(4 + t) - 8t \\ &= -89 - 31t + 19s \end{aligned}$$

Thus, the solution set is

$$\begin{aligned} x_1 &= -89 - 31t + 19s \\ x_2 &= -17 - 4t + 3s \\ x_3 &= s \\ x_4 &= 4 + t \\ x_5 &= t \end{aligned}$$

**where $t$ and $s$ are arbitrary real numbers.**. Choose arbitrary numbers for $t$ and $s$ and substitute the corresponding list $(x_1, x_2, \ldots, x_5)$ into the system of equations to verify that it is a solution. □

## 2.3 Existence and uniqueness of solutions

The REF or RREF of an augmented matrix leads to three distinct possibilities for the solution set of a linear system.

**Theorem 2.5:** Let $[\mathbf{A}\ \mathbf{b}]$ be the augmented matrix of a linear system. One of the following distinct possibilities will occur:

1. The augmented matrix will contain an inconsistent row.
2. All the rows of the augmented matrix are consistent and there are no free parameters.
3. All the rows of the augmented matrix are consistent and there are $d \geq 1$ variables that must be set to arbitrary parameters

In Case 1., the linear system is inconsistent and thus has no solution. In Case 2., the linear system is consistent and has only one (and thus **unique**) solution. This case occurs when $r = \text{rank}(\mathbf{A}) = n$ since then the number of free parameters is $d = n - r = 0$. In Case 3., the linear system is consistent and has infinitely many solutions. This case occurs when $r < n$ and thus $d = n - r > 0$ is the number of free parameters.

**After this lecture you should know the following:**

- what the REF is and how to compute it
- what the RREF is and how to compute it
- how to solve linear systems using row reduction (**Practice!!!**)
- how to identify when a linear system is inconsistent
- how to identify when a linear system is consistent
- what is the rank of a matrix
- how to compute the number of free parameters in a solution set
- what are the three possible cases for the solution set of a linear system (Theorem 2.5)